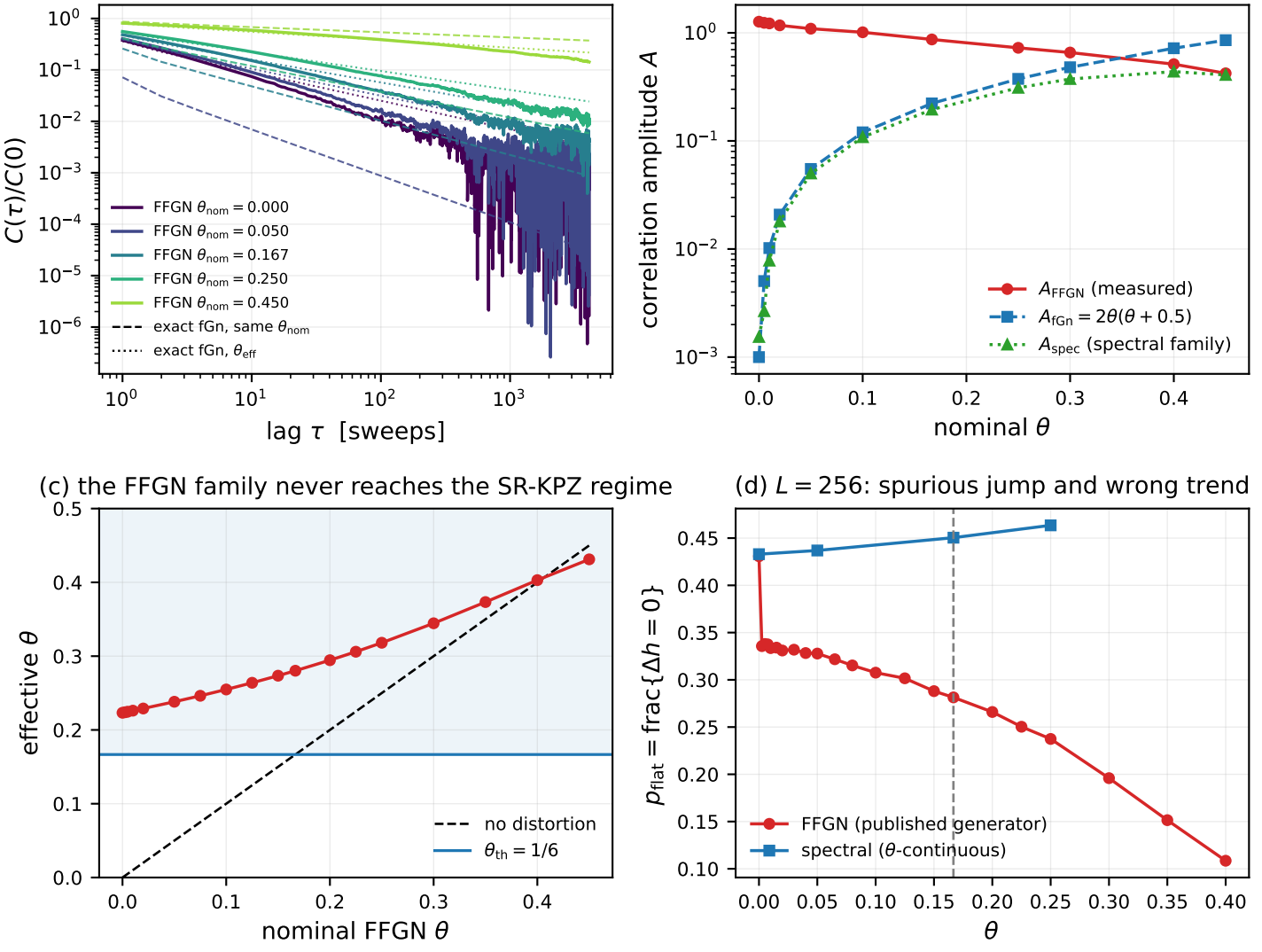

(a) FFGN vs exact fGn
$C(\tau)/C(0)$
lag τ [sweeps]
FFGN $\theta_{\mathrm{nom}} = 0.000$
FFGN $\theta_{\mathrm{nom}} = 0.050$
FFGN $\theta_{\mathrm{nom}} = 0.167$
FFGN $\theta_{\mathrm{nom}} = 0.250$
FFGN $\theta_{\mathrm{nom}} = 0.450$
exact fGn, same θ_{nom}
exact fGn, θ_{eff}
(b) amplitude: FFGN does not vanish at $\theta \to 0$
correlation amplitude A
nominal θ
A_{FFGN} (measured)
$A_{\mathrm{fGn}} = 2\theta(\theta + 0.5)$
A_{spec} (spectral family)
(c) the FFGN family never reaches the SR-KPZ regime
effective θ
nominal FFGN θ
no distortion
$\theta_{\mathrm{th}} = 1/6$
(d) $L = 256$: spurious jump and wrong trend
$p_{\mathrm{flat}} = \mathrm{frac}\{\Delta h = 0\}$
θ
FFGN (published generator)
spectral (θ-continuous)